Exoplanet Detection using Machine Learning Techniques

Mohamad Almasri
Computer Eng. Dept.
PSUT
Amman, Jordan
moh20210608@std.psut.edu.jo

Kinan Radaideh
Computer Eng. Dept.
PSUT
Amman, Jordan
kin20210405@std.psut.edu.jo

Dr. Heba Abdel-Nabi
Computer Eng. Dept.
PSUT
Amman, Jordan
h.abdelnabi@psut.edu.jo

***Abstract*—Exoplanet detection is a crucial domain in modern astronomy, primarily driven by the analysis of transit signals from missions like the Kepler Space Telescope. However, distinguishing true exoplanets from false positives, such as eclipsing binaries, gas and rock formations, or instrumental noise, remains a challenge. This study presents a machine learning approach to automate the validation of Kepler Objects of Interest (KOIs) by applying and evaluating two algorithms: a Decision Tree and a Logistic Regression model. A critical component of this research is the elimination of data leakage; flags that artificially inflated initial model performance (yielding up to 98% accuracy) were removed to force the models to learn exclusively from raw physical and scientific parameters. Optimization was done with Grid Search and 5-Fold Stratified Cross-Validation. The Decision Tree emerged as the better model, achieving an accuracy of 90%, an F1-Score of 0.84, and an AUC of 0.942 on the physical-only dataset. The Logistic Regression model, however, could not model the non-linear nature of astronomical phenomena, achieving only 84% accuracy and suffering from a high false negative rate. These results demonstrate that hierarchical, tree-based machine learning architectures can effectively and accurately classify exoplanetary candidates, providing a framework applicable to current and future astronomical surveys.**

***Keywords*—Exoplanet Detection, Kepler Mission, Machine Learning, Decision Trees, Data Leakage, Logistic Regression, Transit Photometry**

I. INTRODUCTION

Exoplanet detection, the process of discovering planets outside of the Solar System that orbit around a star other than the Sun, has been groundbreaking in the field of planetary studies. One of the methods used in detecting exoplanets involves analyzing the decrease in brightness as a result of a planetary transit across the star. In particular, one of the more successful missions involved the use of NASA's Kepler space telescope, which managed to detect thousands of exoplanets in its survey. But because of various reasons, not all of the transit-type signals that have been detected using this method correspond to exoplanets; several false positives include eclipsing binaries, background systems, and massive gas or meteorite formations.

The purpose of this project is to create a classification model that can effectively discriminate between confirmed exoplanets and false positives using the Kepler Object of Interest (KOI) dataset [1], which consists of 9,564 observations with 50 features (e.g., orbital elements, physical characteristics, stellar properties). Two algorithms were used, optimized from a set of hyperparameters with grid search and k-fold cross validation: the decision tree and the logistic regression algorithms. These two algorithms are then applied and compared.

One of these features, koi_disposition, contains the label for each instance, between FALSE POSITIVE (not an exoplanet), CANDIDATE (uncertain classification) or CONFIRMED (a confirmed exoplanet observation). This is the target variable, which, after removing the CANDIDATE labels, was reduced into a binary classification task. Another important step in data preprocessing is an accurate exclusion of features to avoid data leakage. Several features related to the final decision, like koi_pdisposition, koi_score, and flags_columns (koi_fpflag_*), were excluded to make sure that a machine learning model would use raw scientific measurements to learn from rather than predefined labels. Other irrelevant columns including empty or constant columns were excluded, leaving 15 features in total. After all features which were causing data leakage were removed, as well as all instances with null values, a final, cleaned dataset of 7,015 observations was produced.

During exploratory data analysis, it became obvious that there was a significant discrepancy in the distribution of several features between objects with positive classes. Confirmed planets showed narrower, physically reasonable distributions (smaller planetary radii, lower transit depths, etc.), while false positives demonstrated wide distributions with many extreme outliers. The most prominent cases were shown in boxplots, KDEs, and scatter plots indicating separation on physical basis. This section describes in more details the data preprocessing procedure, feature selection criteria, model training, and results of model evaluation.

II. RELATED WORK

This work [2] used a one dimensional convolutional neural network with max pooling to classify Kepler crossing events as either exoplanets or false positives. The model was trained on light curves and used both a global view of the whole folded light curve and a local view zoomed in on the transit. On the test set, the work found that its model ranked real planet signals above false positives 98.8% of the time. The work then applied the model to a new search of known multi planet systems and discovered two new planets. One of them, Kepler 80 g, became part of a five planet resonant chain. The other, Kepler 90 i, brought the total number of planets around that star to eight, which tied with our Sun as the star with the most known planets

This paper [3] devises a method to indicate whether a signal found in Kepler data looks like a transiting planet or not. The goal was to automatically sort through the thousands of signals the Kepler pipeline finds and extract those which were transits. The work used dimensionality reduction to compress the folded light curves into fewer dimensions, and then used k nearest neighbors to measure how close a signal was to known transits. They called this the LPP transit metric. When tested, this metric was able to remove more than 90% of the non-transit signals while keeping more than 99% of the known planet candidates. They also tested it on fake transits injected into the data and lost less than 1% of them. The work concluded that this metric can help missions such as NASA's Transiting Exoplanet Survey Satellite (TESS) [4] quickly find planet candidates.

This work [5] looked at using machine learning to detect exoplanets from light curves, similar to [1]. The goal was to see if classical machine learning could work well without needing the computational cost of deep learning. The work analyzed light curves and extracted 789 features from each using TSFresh. They then trained a gradient boosting classifier, lightGBM, on those features. When tested on simulated data, the researchers found their method worked better than the conventional least squares method. On real Kepler data, the method was able to distinguish signals from true planets and false positives with an AUC of 0.948. The recall was 0.96. On TESS data, the method got 98% accuracy and a recall of 0.82 at a precision of 0.63. The work concluded that this classical approach is much faster than deep learning but could nonetheless produce good results.

This work [6] used five machine learning algorithms were developed to predict wether a KOI event was an exoplanet or false positive: logistic regression, support vector classifier, gradient boosting, random forest and multilayer perceptron. These five algorithms were applied to fifteen KOI attributes. Among all of the five algorithms tested, the random forest algorithm had the highest level of accuracy with an average accuracy rate of 99% using cross-validation. By examining which attributes contributed most to the accuracy of the prediction, it was determined that approximately seventy-five percent of the attribute contributions came from the Not Transitlike Flag, Centroid Offset Flag, Stellar Eclipse Flag, Ephemeris Match Indicate Contamination Flag and Planetary Radius. The results indicate that by concentrating on these specific KOI attributes, future space missions such as TESS may be able to evaluate exoplanet candidate systems much more quickly than is currently possible.

This paper [7] used the Kepler data set, provided by NASA, and analyzed it with four supervised algorithms (decision tree, random forest, naive Bayes, and artificial neural network), while another set of confirmed exoplanets was analyzed through the application of k-means clustering (unsupervised approach), to predict whether a KOI instance was an exoplanet or not. The results showed that the authors obtained accuracies of 99.06%, 92.11%, 88.50%, and 99.79% respectively, through the use of decision tree, random forest, naive Bayes, and artificial neural network. At the same time, they were able to cluster the exoplanets and discover which of them belong to the same cluster as Earth.

This work [8] used machine learning techniques to automate the classification of Kepler transit signals into three classes – planets, eclipsing binary stars, and non-transiting events. This was accomplished by applying the random forest technique with 77 input variables taken from the Kepler pipeline. The error rate in the overall classification was 5.85% and the error rate in misclassification of planet candidates was 2.81%. It was concluded that random forests could be an effective tool for automating transit signal validation.

III. EXPERIMENTAL SETUP

A. Dataset Description

The data utilized for this research is the Kepler Objects of Interest (KOI) dataset, sourced from NASA's Kepler Space Telescope mission via Kaggle. The Kepler mission was designed to discover Earth-sized exoplanets by continuously monitoring the brightness of over 150,000 stars to detect transits—periodic dimming events caused by planets crossing in front of their host stars. The specific dataset evaluated in this study consists of 9,564 observational instances and 50 initial attributes.

The 50 raw attributes can be broadly categorized into three distinct groups: identification tags (e.g., unique target star tracking numbers), automated pipeline disposition flags (e.g., algorithmic flags indicating background noise or telescope glitches), and critical physical and stellar parameters. These physical parameters mathematically describe the orbital mechanics of the transit and the characteristics of the host star. The latter 2 categories were largely removed from the dataset to prevent data leakage and overfitting, since they were irrelevant to the prediction task. Table I summarizes several of the most influential physical and pipeline features used to evaluate the physical plausibility of a planetary candidate.

TABLE I
REPRESENTATIVE FEATURES FROM THE KEPLER DATASET

Category	Raw Feature	Simplified Description
Physical Planetary	*koi_period*	**exoplanetYearInDays**: Time (days) to orbit the star.
	koi_duration	**transitDuration**: Time (hours) the transit event lasts.
	koi_prad	**planetaryRadiusInEarths**: Planet size relative to Earth.
	koi_depth	**percentageBlockedStarlight**: Light lost during transit.
	koi_teq	**planetTemperature**: Estimated temperature of the planet.
Physical Stellar	*koi_steff*	**starTemperature**: Surface temperature of the host star.
	koi_srad	**starSizeInSuns**: Radius of the host star relative to the Sun.
Flags	*koi_fpflag_nt*	**is_telescopeGlitch**: Flag for telescope instrument errors.
	koi_fpflag_ss	**is_eclipsingBinary**: Flag for a secondary star blocking light.
	koi_fpflag_co	**is_fromBackgroundStar**: Flag for nearby star interference.
Leakage	*koi_score*	Confidence score from the automated pipeline.
	koi_pdisposition	The pipeline's initial automated guess of the class.
IDs	*kepid*	Unique tracking ID for the target star.
	kepoi_name	A tracking string for the specific transit signal.

B. Target Variable and Class Imbalance

The primary goal of the predictive models in this study is to accurately classify the koi_disposition attribute. In the raw Kepler dataset, this target variable has three distinct classifications: CONFIRMED (verified exoplanets), FALSE POSITIVE (non-planetary signals, such as eclipsing binary stars, telescope noise, or gas formations), and CANDIDATE (objects not yet verified or rejected).

To help the machine learning algorithms learn clear physical boundaries, all CANDIDATE instances were removed during data preparation. As a result, the task became a strict binary classification problem.

The remaining data shows a significant class imbalance: 5,023 FALSE POSITIVE instances compared to only 2,293 CONFIRMED exoplanets. This creates roughly a 2:1 ratio in favor of false alarms. This skew reflects real-world astronomical surveys, where background noise and false signals vastly outnumber actual planets.

However, this imbalance presents a challenge for machine learning. Algorithms trained on skewed data often bias their predictions toward the majority class (False Positives). This artificially inflates standard accuracy scores while causing the model to miss the minority class (Confirmed Planets). Because of this, specialized evaluation metrics are needed to properly assess model performance. Additionally, stratified sampling was used throughout the study to help handle this imbalance.

C. Evaluation Metrics

Due to the significant class imbalance within the dataset, relying solely on standard classification accuracy provides an incomplete measure of model performance. In highly skewed datasets, a model can achieve artificially high accuracy simply by predicting the majority class (False Positives) for all instances, while failing entirely to identify the minority class. Therefore, this study evaluates the algorithms using a robust set of metrics derived from the confusion matrix: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

1) Precision and Recall: Precision measures the proportion of positive predictions that are genuinely correct. High precision is critical in astronomical surveys to avoid allocating costly telescope resources to false alarms. Recall (or Sensitivity) measures the proportion of actual exoplanets successfully identified by the model, ensuring true planetary candidates are not overlooked.

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

2) F1-Score: To establish a balance between minimizing false alarms and maximizing discoveries, the primary metric utilized in this study is the F1-Score. This metric calculates the harmonic mean of Precision and Recall, ensuring the model is heavily penalized for poor performance on the minority class.

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

3) Receiver Operating Characteristic (ROC) and AUC: While the F1-Score evaluates performance at a static decision threshold (typically 0.5), the Receiver Operating Characteristic (ROC) curve evaluates the model's diagnostic ability across all possible probability thresholds. The ROC curve plots the True Positive Rate (TPR), mathematically equivalent to Recall, against the False Positive Rate (FPR).

$$TPR = \frac{TP}{TP + FN} \tag{4}$$

$$FPR = \frac{FP}{FP + TN} \tag{5}$$

In this context, the models output a continuous probability score (from 0.0 to 1.0) representing the likelihood an object is a confirmed exoplanet. By sweeping the decision threshold across this range, the ROC curve visually demonstrates the trade-off between maximizing planetary discoveries (high TPR) and minimizing false alarms (low FPR).

To quantify this curve into a single comparative metric, the Area Under the ROC Curve (AUC) is calculated. A model performing random guesses yields an AUC of 0.5, whereas a model that perfectly separates the classes achieves an AUC of 1.0. Because AUC measures the overall discriminative power of the model independent of a specific threshold, it serves as the primary metric to compare different algorithm architectures (such as Decision Trees versus Logistic Regression) without the results being skewed by the dataset's 2:1 class imbalance.

D. Development Environment

All computational experiments, data processing, and predictive modeling were conducted using Python 3 within a Google Colab environment. Several openly available libraries were used for the completion of the project.

Data manipulation and numerical calculations, such as filtering classes, managing missing values, and creating and manipulating features, were handled using the Pandas and NumPy libraries. To perform Exploratory Data Analysis (EDA) and data visualization, Matplotlib and Seaborn were used. The machine learning phase, including model construction, hyperparameter tuning via cross-validation, and metric evaluation, was implemented using the Scikit-Learn framework.

IV. ALGORITHM

A. Data Preparation and Cleaning

To ensure the machine learning algorithms could effectively identify underlying physical patterns, the raw Kepler dataset underwent a preparation and cleaning phase.

1) Target Variable Encoding and Class Filtering: The target variable, koi_disposition, originally contained three categories. Since CANDIDATE represented an uncertain class, a KOI for which the classification was unclear, it was removed, since the models require concrete classes. Two classes remained, reducing the problem to a binary classification problem. These remaining classes were then numerically encoded, assigning a value of 1 to CONFIRMED exoplanets and 0 to FALSE POSITIVE instances.

2) Feature Renaming and Missing Values: To improve the readability of the dataset, features with technical shorthand names set by NASA were renamed with intuitive physical parameter names (e.g., converting koi_prad to planetaryRadiusInEarths). Following this, the dataset was analyzed for missing values. Columns that were entirely empty, such as specific temperature error margins, were dropped completely, and any remaining rows with missing data were removed entirely rather than being imputed because physical data collected during astronomical missions already have inherent uncertainty, and imputing would not have been beneficial for physical data.

3) Feature Elimination and Leakage Prevention: A critical step in the preparation phase was the removal of features that provided no physical predictive power or posed a risk of data leakage. First, administrative identification tags (e.g., kepid) were dropped as they carry no astronomical significance. Second, all scientific error margin columns (e.g., koi_period_err1) were removed, as these represent telescope measurement uncertainties rather than the actual physical characteristics of the astronomical bodies. Finally, automated pipeline flags (e.g.,koi_fpflag_nt) and pipeline predisposition scores, which were scores set by individual astronomers estimating the confidence of instance classification, were discarded. Because these flags are generated by NASA's analysis teams to specifically identify false positives, retaining them would result in data leakage. Removing them forces the models to learn exclusively from raw physical parameters (such as planetary size and stellar radius) rather than relying on prior conclusions.

4) Data Splitting and Scaling: Because the dataset contains variables with vastly different magnitudes, such as stellar temperatures in thousands of Kelvin alongside planetary radii measured in single digits, feature scaling was required. First, the dataset was partitioned into training and testing sets using an 80/20 split. Following the split, a StandardScaler was applied to the training data to normalize the features (centering the data to a mean of zero and a standard deviation of one), with the same transformation then applied to the test set. Much of the underlying features were already normally distributed, so this standardization did not radically transform the data in a way that was misrepresentative.

B. Exploratory Data Analysis and Feature Selection

The initial phase of the EDA identified the presence of columns with 100% missing values (such as specific error margins), which informed the null-removal steps during data preparation.

1) Target Variable Distribution: The distribution of the target variable, *Disposition*, was evaluated to understand the baseline probability. As illustrated in Fig. 1, after filtering out the ambiguous candidate class, the dataset showed a clear class imbalance, containing 5,023 *FALSE POSITIVE* instances and 2,293 *CONFIRMED* instances. In later steps, therefore, data was sampled using stratified sampling to remedy this.

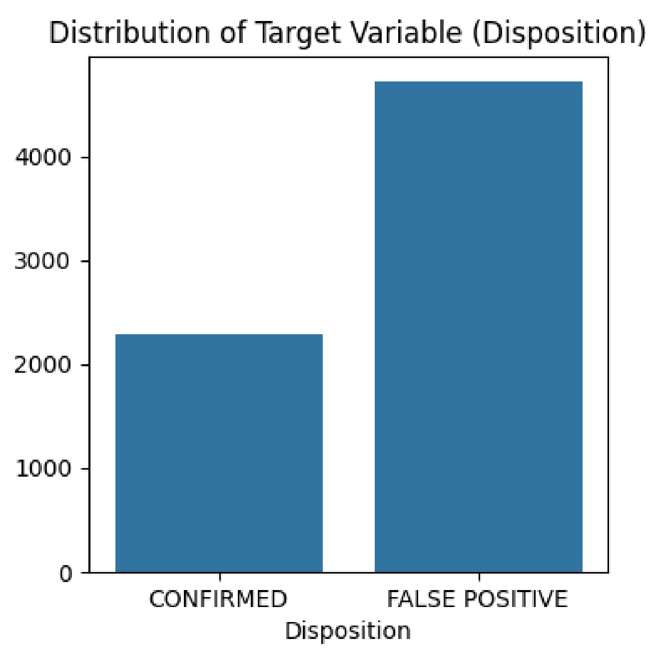

Fig. 1. Class distribution of the *Disposition* variable, highlighting the roughly 2:1 imbalance between False Positives and Confirmed exoplanets.

2) Descriptive Statistics: Standard descriptive statistics were generated for some of the cleaned features. Table II provides a summarized excerpt of these statistics for key astronomical parameters. The vast difference in magnitudes, ranging from percentages (*percentageBlockedStarlight*) to thousands of units (*exoplanetYearInDays*), mathematically justifies the necessity of the *StandardScaler* applied during preprocessing.

TABLE II
SUMMARY STATISTICS FOR KEY PHYSICAL FEATURES

Feature	Mean	Std. Dev.	Min	Max
exoplanetYearInDays	46.12	109.2	0.25	1071.2
transitDuration	5.11	4.2	0.16	138.5
percentageBlockedStarlight	22010	79040	1.0	1541400
planetaryRadiusInEarths	101.4	3192	0.14	200346

3) Univariate Analysis (Boxplots and KDEs): To visualize the spread of the data, logarithmic box plots (Fig. 2) and Kernel Density Estimation (KDE) plots (Fig. 3) were generated. The boxplots revealed extreme upper-bound outliers, especially in the False Positive class across almost all physical metrics.

The KDE plots further illuminated this phenomenon. For the *CONFIRMED* class, core physical features such as *planetaryRadiusInEarths* and *percentageBlockedStarlight* have approximately normal (Gaussian) distributions, which further justify standardization scaling, centered at highly localized, physically plausible values. In contrast, the distributions for the *FALSE POSITIVE* class are often multimodal and spread across massive ranges. This visualizes the nature of astronomical false positives: while true planets are constrained by the laws of physics to normal distributions of size and orbit, false positives are triggered by a mixture of massive eclipsing binary stars, gas and rock formations, and mechanical telescope glitches, resulting in extreme variance. In these and other graphs, logarithmic axes were used because of the huge range of values taken on by KOI instances.

4) Bivariate and Correlation Analysis: A correlation matrix (Fig. 4) and corresponding scatter plots (Fig. 5) were developed to help understand interactions between features. Overall, interactions between features were weak; which is a positive sign since it means our dataset's sampling is general and representative, in other words that there is no sampling bias. In other cases, the chosen features describe underlying physics and astronomy well. For example, plotting *Solar Radiation (in Earths)* against *Planet Surface Temp (K)* shows a perfect curve, reflecting the existing physical relationship between them. A considerable correlation between *percentageBlockedStarlight* against *planetaryRadiusInEarths* is also revealed, while the remaining features are less correlated.

5) Programmatic Feature Selection (Decision Trees): An algorithmic feature selection approach, which involves collecting feature importance, was used, by extracting the Gini importance from the baseline Decision Tree classifier.

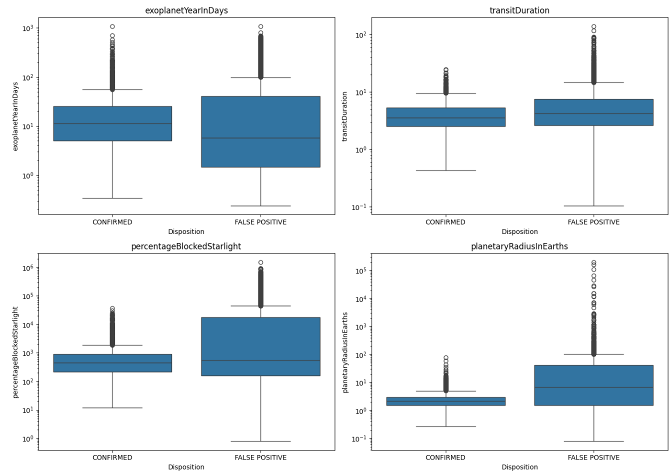

Fig. 2. Logarithmic box plots demonstrating the extreme variance and outliers present in the False Positive class.

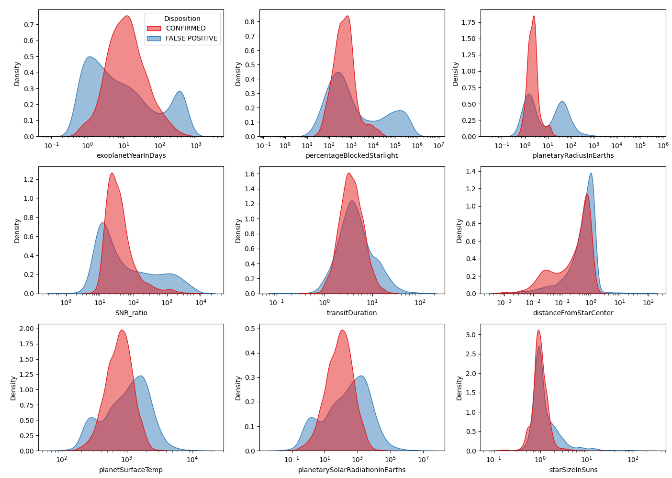

Fig. 3. KDE distributions revealing normal distributions for confirmed planets and multimodal variance for false alarms.

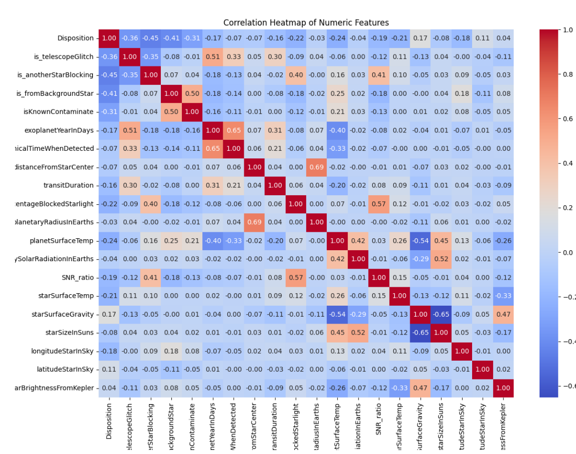

Fig. 4. Correlation heatmap identifying instances of multicollinearity among physical stellar parameters.

Fig. 5. Bivariate scatter plots illustrating perfect physical dependencies, reasonable correlations, and uncorrelated astronomical noise.

Initially, a tree was trained on a dataset that included the NASA pipeline flags (Fig. 6). The resulting graph showed that non-physical flags overwhelmingly dominated feature importance. This dominance proved that the flags were causing data leakage, allowing the model to bypass actual physics; predicting based solely on flags set by astronomers rather than from physical features such as planetary year in Earth days and such. This justified the removal of these flags (*is_telescopeGlitch*, *is_fromBackgroundStarm*, *isKnownContaminate*, *is_anotherStarBlocking*) during the data cleaning phase.

A second Decision Tree was then trained using only the physical parameters (Fig. 7). The model identified *SNR_ratio* and *planetaryRadiusInEarths* as the most critical features. To further simplify the dataset, a selection threshold was set to remove any feature contributing less than 0.1% to the model's decision-making process. Interestingly, when this threshold was applied to the physics-only dataset, none of the features were removed. This finding proved that unlike the flag-dominated dataset, every remaining astronomical parameter contributes at least marginal, valid predictive power toward the final classification boundary.

C. Predictive Modeling Algorithms and Optimization

Two machine learning algorithms were implemented and evaluated: a Decision Tree classifier and Logistic Regression. To quantify the impact of data leakage, the entire training, optimization, and evaluation pipeline was executed twice for both models: initially including the NASA pipeline flags (e.g., *is_telescopeGlitch* and *is_fromBackgroundStar*) and then excluding them to force the models to rely on physical parameters, as would be more practical. The rest of the analysis relied on these flag-free models, with which the error metrics were calculated. Model performance during these training phases was assessed using the previously discussed error metrics, the F1-Score and ROC-AUC.

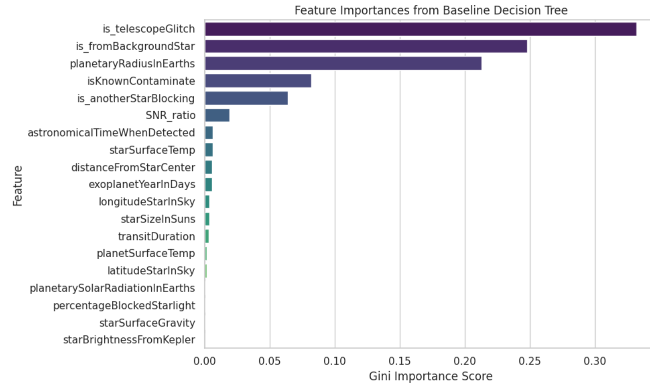

Fig. 6. Initial feature importance demonstrating the dominating effect of data-leakage pipeline flags.

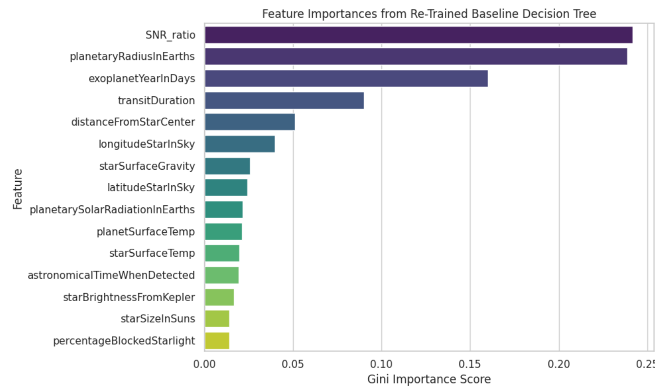

Fig. 7. Final feature importance strictly evaluating physical parameters, where all remaining features exceeded the 0.1% utility threshold.

1) Decision Tree Classifier: The modeling phase began with a Decision Tree.

Unlike parametric models, a Decision Tree makes no assumptions about the underlying distribution of the data. It operates by recursively dividing the feature space into hierarchical nodes, splitting the training data at each step based on the feature threshold that maximizes information gain and minimizes impurity. This non-linear, hierarchical structure is well-suited for the KOI dataset; it adjusts for the interactions between variables (such as the relationship between *planetaryRadiusInEarths* and *SNR_ratio*) and mimics the sequential vetting logic employed by human astronomers when verifying Objects of Interest.

To prevent the algorithm from creating infinitely deep branches that overfit, hyperparameter tuning was conducted using a Grid Search combined with 5-Fold Stratified Cross-Validation. The predefined search space (the grid) optimized several structural constraints:

- *criterion*: The mathematical function utilized to measure the quality of a split (testing both *gini* impurity and *entropy* for information gain).
- *max_depth*: The maximum permitted number of levels in the tree, acting as a hard cap on model complexity.

- *min_samples_split*: The minimum number of observations required in a node before the algorithm is allowed to split it further.
- *min_samples_leaf*: The minimum number of observations required to exist at a final leaf node, preventing the creation of highly specific rules based on single outliers.

2) Logistic Regression: Following the non-linear tree model, Logistic Regression was implemented as a parametric, linear baseline.

This algorithm operates by applying a logistic (sigmoid) function to a linear combination of the scaled input features, outputting a probability score bounded between 0 and 1. This score represents the mathematical likelihood that a specific instance belongs to the *CONFIRMED* disposition. As a highly interpretable model, it allows for clear observation of how individual scaled features linearly impact the final prediction.

Similar to the Decision Tree, the Logistic Regression model was optimized using GridSearchCV alongside 5-Fold Stratified Cross-Validation. The search space included:

- *C*: The inverse of the regularization strength. Smaller values of C specify stronger regularization, penalizing the model for assigning excessive weight to any single physical feature, preventing overfitting.
- *penalty*: The specific regularization norm applied (testing both *L1* and *L2* penalties) to manage feature weights and handle slightly correlated astronomical parameters.
- *solver*: The algorithm used for optimization (such as *liblinear* or *lbfgs*; in the case of this paper the latter), selected to ensure efficient convergence on the scaled dataset.

V. RESULTS & DISCUSSION

A. Optimized Model Hyperparameters

Table III describes the parameters and hyperpameters of the resulting models after training and application. Four models are shown: the Decision Tree and Logistic Regression models, both with and without the data-leaking flags.

TABLE III
OPTIMAL HYPERPARAMETERS IDENTIFIED VIA GRIDSEARCHCV

Model & Configuration	Tuned Hyperparameters
Decision Tree (Physics Only)	*criterion*: gini *max_depth*: None *min_samples_leaf*: 20 *min_samples_split*: 2
Logistic Regression(Physics Only)	*C*: 1000 *max_iter*: 2000 *solver*: lbfgs *penalty*: l2 (Default)
Decision Tree (With Flags)	*criterion*: gini *max_depth*: None *min_samples_leaf*: 20 *min_samples_split*: 2
Logistic Regression (With Flags)	*C*: 10 *max_iter*: 2000 *solver*: lbfgs *penalty*: l2 (Default)

B. Impact of Data Leakage

To demonstrate the impact of data leakage, the models were initially trained on a dataset that retained the automated NASA vetting flags (e.g., *is_telescopeGlitch* and *is_fromBackgroundStar*). With these flags, both the Logistic Regression and Decision Tree models achieved an artificially inflated classification accuracy of 98.0%. While numerically impressive, feature importance extraction confirmed that the algorithms were almost entirely ignoring the physical properties of the celestial bodies (such as planetary radius and transit depth) and instead basing their predictions solely on the algorithmic flags previously assigned by the NASA astronomers. This baseline execution proved that retaining these features fundamentally invalidates the scientific utility of the machine learning task, since a prediction is only useful if it is based on fundamental physical variables. Thus, all subsequent results reported in this study are from the models' evaluations after these leakage-inducing flags were removed from the feature list.

C. Predictive Performance on Physical Parameters

After the removal of the data-leaking flags, the algorithms were forced to classify instances relying on physical and astronomical parameters. 15 such features remained to be used by the models. Table IV presents the final comparative performance of both the Decision Tree and Logistic Regression models evaluated on the unseen testing data.

TABLE IV
CLASSIFICATION PERFORMANCE ON PHYSICAL PARAMETERS

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	**0.90**	**0.86**	**0.82**	**0.84**
Logistic Regression	0.84	0.74	0.78	0.76

The optimized Decision Tree classifier emerged as the superior model, achieving an overall accuracy of 90% and an F1-Score of 0.84 for the *CONFIRMED* planet class. Notably, the Decision Tree maintained a strong balance between Precision (0.86) and Recall (0.82), thus producing a higher F1 score, indicating it was highly capable of minimizing false astronomical alarms, despite them representing the majority of the instances in the dataset, and at the same successfully identifying the vast majority of true planetary signals. The Logistic Regression model performed more poorly when restricted to physical parameters, achieving a lower overall accuracy of 84%. While its Recall remained relatively operable (0.78), its Precision dropped significantly to 0.74, resulting in a lower F1-Score of 0.76. This drop in Precision indicates that the Logistic Regression model suffered from a higher rate of false positives, misclassifying background noise or false positives such as eclipsing binaries as confirmed exoplanets.

This performance gap mathematically supports the hypothesis that the physical characteristics separating exoplanets from false positives (such as the relationship between transit

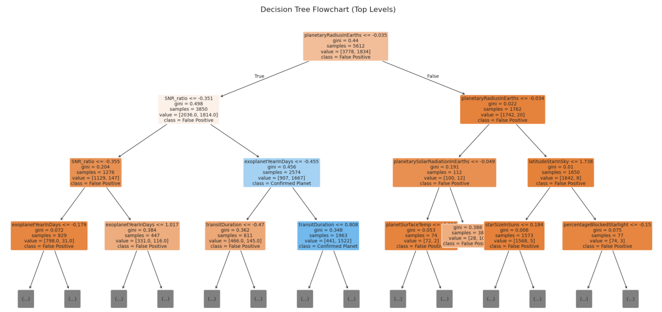

Fig. 8. Visualization of the chosen, applied Decision Tree. The tree in its true form has 18 layers in total. The figure shows its hierarchical logic, how it applies non-linear thresholds to astronomical parameters to make predictions.

depth, planetary radius, and signal-to-noise ratio) are fundamentally non-linear, which Decision Trees are more capable of handling, whereas the linear assumptions of the Logistic Regression model proved insufficient for the full complexity of the KOI dataset.

D. Further Error Analysis and Performance Visualization

More error analysis techniques, including visual ones, were used to assess the models.

1) Confusion Matrix Analysis: A confusion matrix classifies predictions between True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). As illustrated in Fig. 9, the confusion matrix reveals where the Logistic Regression model failed. Compared to the Decision Tree, the linear model produced a higher count of False Negatives. This highlights the drop in Recall previously noted; the linear model was too rigid, misclassifying true planetary signals as background noise. The Decision Tree minimized both error types much more effectively.

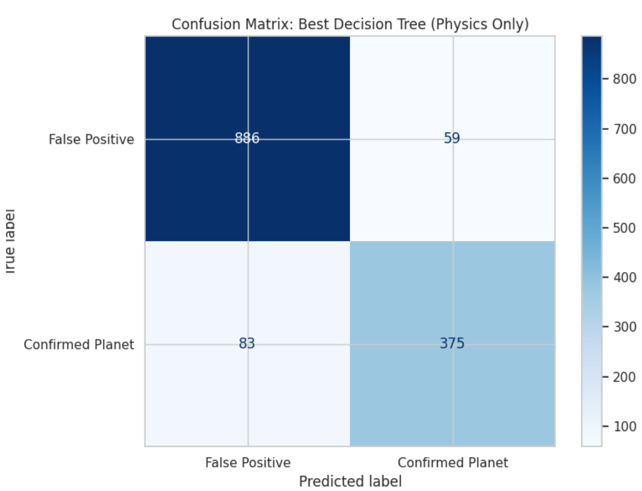

Fig. 9. Confusion matrices comparing the exact classification errors. The Decision Tree demonstrates a superior balance, while Logistic Regression shows a vulnerability to False Negatives.

2) Receiver Operating Characteristic (ROC) Curve: The ROC curve illustrates the predictive ability of the models by plotting the True Positive Rate (also called Recall) against the False Positive Rate across all possible thresholds. An ideal model's curve will be perfectly parallel with the y-axis, indicating it can maximize planetary discoveries while keeping false alarms zero.

Now that the Decision Tree has been established as being the superior model, its performance is more closely scrutinized. As shown in Fig. 10, the Decision Tree maintains a significantly large Area Under the Curve (AUC). This visualization shows that regardless of how strictly the probability threshold is set, the non-linear tree architecture is at distinguishing between the physical parameters of the two classes of Disposition.

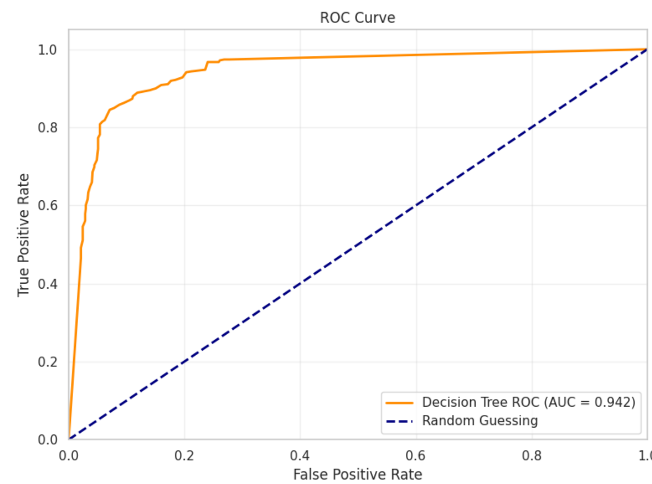

Fig. 10. ROC Curves demonstrating the Decision Tree's ability to maximize true positive discoveries across all probability thresholds.

Overview of Results: Overall, these visualizations provide more evidence of the superiority of the Decision Tree relative to the Logistic Regression model. The physical parameters governing planetary transits, and the distinction between planets and non-planets, are simply linear. The overlap between genuine exoplanets and background astronomical anomalies requires a model capable of modeling highly non-linear relationships. The hierarchical, threshold-based and non-linear logic of the Decision Tree proved highly effective for this task, outperforming the linear assumptions of Logistic Regression.

E. Comparison with Other Papers

This comparison focuses on two primary aspects: predictive accuracy, and the trade-off between feature complexity and scientific interpretability.

As detailed in Table V, the 90% accuracy achieved by the Decision Tree in this study compares favorably with established research, particularly when considering the deliberate removal of data-leaking flags.

TABLE V
COMPARATIVE ANALYSIS WITH PRIOR RESEARCH

Study	Algorithm	Features	Results
Malik et al. [5]	LightGBM	789	0.948 AUC
Srivathsa et al. [6]	Random Forest	15 (Flags)	99.0% Acc
This Study	**Decision Tree**	**15 (Physics)**	**0.942 AUC**

1) Interpretability and Data Leakage: The work [6] reported a high classification accuracy of 99% using a Random Forest model on 15 features. However, that study noted that approximately 75% of the model's predictive power was derived from NASA's automated vetting flags, such as the *Centroid Offset* and *Ephemeris Match* flags. While this study's Decision Tree achieved a lower absolute accuracy of 90%, it did so while relying strictly on raw physical parameters. This suggests that the model used in this paper is more scientifically robust for the task in surveys where such flags have not yet been generated by human analysts.

2) Model Efficiency and AUC: Malik et al. [5] used a Gradient Boosting classifier (LightGBM) and extensive feature engineering (789 features) to achieve an AUC of 0.948 on Kepler data. In comparison, the optimized Decision Tree here achieved a highly similar AUC of 0.942 while utilizing only 15 core astronomical parameters. This comparison shows that a simpler, hierarchical tree-based model can achieve exceptional performance with significantly reduced computational complexity and improved model transparency.

These comparisons show that by focusing on non-linear physical interactions rather than algorithmic metadata, our study provides a viable and efficient framework for automated exoplanet validation.

VI. CONCLUSION

This study successfully demonstrated the use of machine learning in automating the classification of exoplanetary transit signals. By cleaning the KOI dataset and eliminating flags, the models were forced to evaluate candidates based strictly on their underlying physical and astronomical properties. The Decision Tree classifier outperformed the Logistic Regression baseline, achieving an overall accuracy of 90% and an Area Under the Curve (AUC) of 0.942. The non-linear structure of the Decision Tree proved effective for mapping the complex boundaries that separate true exoplanets from background noise and false positives. Ultimately, this research supports the use of machine learning pipelines to effectively identify exoplanets with high precision and recall. Such models hold significant promise for accelerating the discovery process in ongoing and future astronomical missions, ensuring that valuable resources are dedicated to the most scientifically viable planetary candidates.

REFERENCES

[1] NASA, "Kepler exoplanet search results," https://www.kaggle.com/datasets/nasa/kepler-exoplanet-search-results/data, accessed: 2026-05-14.

[2] C. J. Shallue and A. Vanderburg, "Identifying exoplanets with deep learning: A five-planet resonant chain around Kepler-80 and an eighth planet around Kepler-90," *The Astronomical Journal*, vol. 155, no. 1, p. 94, 2018.

[3] S. E. Thompson, F. Mullally, J. Coughlin, J. L. Christiansen, C. E. Henze, M. R. Haas, and C. J. Burke, "A machine learning technique to identify transit shaped signals," *The Astrophysical Journal*, vol. 812, no. 1, p. 46, 2015.

[4] Massachusetts Institute of Technology, "TESS: Transiting exoplanet survey satellite," https://tess.mit.edu/, accessed: 2026-05-14.

[5] A. Malik, B. P. Moster, and C. Obermeier, "Exoplanet detection using machine learning," *arXiv preprint arXiv:2011.14135*, 2020.

[6] V. Srivathsa and R. Assaf, "Using machine learning to determine the most important features in exoplanet verification," *Journal of Student Research*, vol. 11, no. 3, 2022.

[7] Y. Jin, L. Yang, and C. E. Chiang, "Identifying exoplanets with machine learning methods: A preliminary study," *International Journal on Cybernetics & Informatics*, vol. 11, no. 1/2, pp. 31–42, 2022.

[8] S. D. McCauliff, J. M. Jenkins, J. Catanzarite, C. J. Burke, J. L. Coughlin, J. D. Twicken, P. Tenenbaum, S. Seader, J. Li, and M. Cote, "Automatic classification of Kepler planetary transit candidates," *The Astrophysical Journal*, vol. 806, no. 1, p. 6, 2015.